John Doe

Machine Learning Researcher • Software Engineer
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Groningen, Netherlands

Summary

Machine learning researcher and software engineer with a PhD in Neuromorphic Computing and hands-on experience developing deep learning systems in PyTorch. Experienced in end-to-end ML experimentation, Transformer models, neural network optimization, and production software, including contributions integrated into deployed ML stack. Interested in building efficient, reliable ML/DL systems for real-world applications.

Skills

- **Programming Languages:** Python, C++, C
- **Frameworks:** PyTorch, TensorFlow, Scikit-Learn, Hugging Face Transformers
- **Tools:** Git, Anaconda, NNI, Docker
- **Techniques:** Machine Learning, Deep Learning, Model Deployment, Experiment Tracking, Statistical Analysis

Experience

Postdoctoral Researcher (Researcher 4)

University of Groningen repository
Conducted research on Transformers applied to Imitation Learning. Jan 2026 – Sep 2026

- Designed and implemented an end-to-end PyTorch pipeline for a decoder-only Transformer trained through imitation learning from an Actor–Critic expert.
- Developed data-generation, model-training, evaluation, and hyperparameter-optimization workflows, including automated tuning with NNI and reproducible experiment management.
- Evaluated Transformer architectures and inference strategies for sequential decision making, achieving performance comparable to the expert policy on unseen environments.

Intern, SynSense

SynSense repository
Rewrote and extended a PyTorch-based API for a CNN processor. Apr 2024 – Oct 2024

- Implemented and integrated support for non-sequential CNNs with residual connections, enabling reproducible and deployable deep learning models on neuromorphic hardware.
- Contribution merged into the company's **production branch**, becoming part of SynSense's deployed software stack.
- Collaborated with senior engineers through code reviews, testing, and production integration.
- Improved model flexibility and software–hardware compatibility for model deployment.

Full-Stack Developer

AcheiFarmacias repository
Backend, frontend, and mobile app developer for online marketplace. Jul 2018 – Mar 2020

- Built scalable APIs and optimized database queries.
- Engineered mobile app features using Flutter.

Education

Ph.D. in Neuromorphic Computing

University of Groningen Sep 2020 – Dec 2025

Focus on spiking neural networks, algorithm/hardware co-design, and energy-efficient systems.

- Developed and optimized recurrent/spiking neural networks using Hebbian and gradient-based learning.
- Conducted algorithm-hardware co-design for neural networks targeting asynchronous mixed-signal ASICs.
- Published peer-reviewed research on learning algorithms, recurrent neural networks, and neuromorphic hardware.

Master's in Computer Science

Fluminense Federal University Feb 2018 – Feb 2020

Specialized in optimization algorithms with applications in wireless sensor networks. Developed and documented solutions demonstrating strong algorithmic and coding skills.

Bachelor's in Computer Science

Amazonas Federal University Feb 2012 – Dec 2017

Exchange at Fontbonne University (USA) under the Science Without Borders program.

Professional Certifications

Deep Learning Specialization

Coursera certificate

Offered by DeepLearning.AI, tutored by Andrew Ng. Feb 2024 – Jan 2025

- Topics: Deep Neural Networks, Convolutional Neural Networks, Sequence Models, Transformers, Optimization Techniques, and Transfer Learning.

Machine Learning Specialization

Coursera certificate

Offered by Stanford University & DeepLearning.AI, tutored by Andrew Ng. Nov 2023 – Jan 2024

- Topics: Supervised Learning, Unsupervised Learning, Reinforcement Learning, PCA, and Deep Q-Networks.

Publications

Learning in Spiking Neural Networks with a Calcium-based Hebbian Rule for Spike-timing-dependent Plasticity
Phasic Attractors for Flexible and Adaptive Working Memory in Spiking Recurrent Neural Networks
A neuromorphic processor with on-chip learning for beyond-CMOS device integration
Synaptic Normalization for On-Chip Learning in Analog CMOS Spiking Neural Networks
SOFiA: An Algorithm Based on the Dynamics of the Opinion Formation Process

Languages

- **English:** Fluent
- **Portuguese:** Native
- **Japanese:** Beginner (A2)
- **Dutch:** Beginner (A2)